A Theory of International Boycotts

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PSE Globalization and Consumer Preferences Workshop, May 23, 2024

Motivation

Model

Other Aspects of Boycotts

Conclusion


BOYCOTT
NO SEE HEAR BUY
Nestle

Motivation

- **Geoeconomics:** International boycotts are significant in international trade and geopolitical relations.
- **Sanctions?** Unlike government-imposed sanctions and tariffs, collective choice of a collective of consumers
- Key Questions
  - How do international boycotts impact firms and nations?
  - What distinguishes the effects of boycotts from those of sanctions or tariffs?
  - What are the efficiency and equity considerations in boycotts?

Objective

- This paper studies the economics of international boycotts within a **neoclassical Ricardian** framework.
- Studies optimal targeted boycott.
- Key Findings
  - Boycotts can operate on the consumption side or labor supply side
  - **Consumption side:** Boycotts can be more effective as ban on goods near comparative advantage barrier.
  - **Labor supply side:** Boycotters optimally work more to spend on domestic goods
  - Some **symmetry** btw boycotts and VERs but **leakage** from nonboycotters.

Facts on the Impact of Consumer Boycotts

Long history of boycotts.

- With globalization, **"surrogate"** boycotts intensify, citizens target foreign firms due to mainly geopolitical tensions
  - **US Boycott of French wine (Early 2000s)** due to France's opposition to the Iraq war, resulting in a \$112 million sales decrease.
  - **Boycotts of Danish (2015) and French (2015) products** by Arab league.
- Government branches sometimes support consumer boycotts (Bohman 2022, Hong et al. 2011)
  - **Chinese boycott of French cars (2008)** for the support expressed for Tibet, resulting in a loss of market share between 25% and 33%
  - **Boycotts following statements opposing the Chinese government**, which in 50% of cases cause targets to publicly apologize.

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Model Overview

- Theoretical framework based on the Dornbusch, Fischer, and Samuelson (1977).
- Two countries, Home (H) and Foreign (F).
- Trade occurs over a continuum of goods indexed by $j \in [0, 1]$.
- Production
  - Single input: labor (l), with constant marginal product for each good j.
  - Country-specific productivity: $z_H(j)$ for Home and $z_F(j)$ for Foreign.
  - Relative productivity of Home to Foreign: $Z(j) = \frac{z_H(j)}{z_F(j)}$.
  - Labor endowments: l_H and l_F, with free labor movement within countries.

Regular Consumers

Consider regular consumers in both countries:

$$U(\mathbf{q}) = \int_0^1 s(j)\log(q^i(j))dj \tag{1}$$

Constraints:

$$\int_0^1 p(j)q^i(j)dj = w_i l_i \tag{2}$$

- $q^i(j)$: quantity of good j consumed.
- $s(j)$: expenditure share on good j, $S(1) = 1$.
- Non-substitution theorem:
  - prices uniquely determined by supply side $p(j; \{w_H, w_F\})$
  - utilities are determined by vector of world wages $U^H(\{w_H, w_F\})$ and $U^F(\{w_H, w_F\})$

Consumer Boycotters

Share λ of domestic boycotters hold fixed excess demand for domestic and Foreign labor at some values $\left(\overline{L}_H, \overline{L}_F\right) \in [0, l_H] \times [0, l_F]$.

$$U^B(\{w_H, w_F, \overline{L}_H, \overline{L}_F\}) = \max_{(\mathbf{q}_H^B, \mathbf{q}_F^B) \in X} \int_0^1 s(j) \log(q_H^B(j) + q_F^B(j)) dj \tag{3}$$

$$\text{s.t:} \quad \int_0^1 p_H(j) q_H^B(j) dj + \int_0^1 p_F(j) q_F^B(j) dj = w_H l_H \tag{4}$$

$$X = \left\{ \left(\mathbf{q}_H^B, \mathbf{q}_F^B\right) : \quad \left(l_H - \lambda \int_0^1 \frac{q_H^B(j)}{z_H(j)} \, dj, l_F - \lambda \int_0^1 \frac{q_F^B(j)}{z_F(j)} \, dj \right) = \left(\overline{L}_H, \overline{L}_F\right) \right\} \tag{5}$$

Competitive Equilibrium

Definition (Competitive Equilibrium with Boycotters that Hold Fixed Excess Demand)

A CE with boycotters corresponds to a pair of wages, $\{w_H, w_F\}$, a schedule of world prices $\mathbf{p}$, and domestic prices $\{\mathbf{p}_H, \mathbf{p}_F\}$, consumption schedules $\{\mathbf{q}^H, \mathbf{q}^F, \mathbf{q}^B\}$, excess demand for labor inputs $(\overline{L}_H, \overline{L}_F)$, and output schedules that satisfy:

1. Firm optimization: price equal firms' marginal cost,
2. Regular consumer utility maximization: domestic and regular consumers maximize (1) subject to (2),
3. Consumer boycotter optimization: domestic boycotters maximize (3) subject to (4), and (5).
4. Labor market clearing in each country

$$\int_0^1 \frac{((1-\lambda)q^H(j) + q^F(j))}{z_H(j)}\, dj = l_H - \lambda \int_0^1 \frac{q_H^B(j)}{z_H(j)}\, dj = \overline{L}_H, \tag{6}$$

$$\int_0^1 \frac{((1-\lambda)q^H(j) + q^F(j))}{z_F(j)}\, dj = l_F - \lambda \int_0^1 \frac{q_F^B(j)}{z_F(j)}\, dj = \overline{L}_F. \tag{7}$$

Boycotting Coalition Problem

Definition (Boycotting Coalition's Problem)

The boycotting coalition's problem is

$$\max_{(\overline{L}_H,\overline{L}_F)} U^B(\{w_H, w_F, \overline{L}_H, \overline{L}_F\}) \tag{8}$$

$$\text{s.th: } U^F(\{w_H, w_F\}) \leq v^F \tag{9}$$

and conditions (1)-(7) hold.

Expenditure Choices

Denote $E(\mathbf{q}_i^B) \equiv \lambda \int_0^1 p_i(j) q_i^B(j) dj$, for $i \in \{H, F\}$ total expenditure of boycotters on H and F goods

Definition (Competitive Equilibrium with Boycotters with Fixed Expenditures)

A CE with boycotters with fixed expenditures corresponds to a pair of wages, $\{w_H, w_F\}$, a schedule of world prices $\mathbf{p}$, and domestic prices $\{\mathbf{p}_H, \mathbf{p}_F\}$, consumption schedules $\{\mathbf{q}^H, \mathbf{q}^F, \mathbf{q}^B\}$, expenditures $\left(\overline{E}_H, \overline{E}_F\right)$, and output schedules that satisfy from optimization, regular consumer utility maximization (1), (2), labor market clearing in each country (6), (7), and consumer boycotter optimization (3) (4), and (10) where the fixed excess demand choice set is replaced by the fixed expenditure for each country goods choice set

$$X = \left\{ \left(\mathbf{q}_H^B, \mathbf{q}_F^B\right) : \quad \left(E(\mathbf{q}_H^B), E(\mathbf{q}_F^B)\right) = \left(\overline{E}_H, \overline{E}_F\right) \right\}. \tag{10}$$

An Equivalence Result

Proposition (Equivalence of CE w Fixed Excess Demand to CE w Fixed Expenditure)

A tuple $\{w_h, w_F, \mathbf{p}, \mathbf{p}_H, \mathbf{p}_F, \mathbf{q}_H, \mathbf{q}_F, \mathbf{q}_B, \overline{L}_H, \overline{L}_F\}$ *is a CE with boycotters with fixed excess demands if and only if* $\{w_h, w_F, \mathbf{p}, \mathbf{p}_H, \mathbf{p}_F, \mathbf{q}_H, \mathbf{q}_F, \mathbf{q}_B, e\left(\overline{L}_H\right), e\left(\overline{L}_F\right)\}$ *is a CE with boycotters with fixed expenditures, with* $e\left(\overline{L}_i\right) = w_i\left(l_i - \overline{L}_i\right)$ *for* $i \in \{H, F\}$.

- How can boycotters allocate their consumption to achieve their goal?
- How do boycotters optimally choose labor supply?

The Consumption Side of Boycotts

Let $S(\mathbf{q}_i^B) \equiv \int_0^1 p_i(j) q_i^B(j) dj / (w_H l_H)$ be total expenditure as share of income of a boycotter

Lemma (Set of Boycotted Goods)

In any CE with boycotters where $\overline{L}_H < l_H$ or $e(\overline{L}_H) > 0$, the set of completely boycotted Foreign goods $\mathcal{J}_B = \{j \in [0,1]; q_F^B(j) = 0\}$ is nonempty.

Proposition (Optimal Set of Boycotted Goods)

Let $\mathcal{J}_B$ be the optimal targeted set of boycotted goods given a level of tolerated Foreign welfare v^F:

1. *There exists no set of positive measure between $\mathcal{J}_B$ and the trade cutoff $\bar{j}$, and $\mathcal{J}_B$ is a convex set.*
2. *There exists $j_B \in [\bar{j}; 1]$ so that $\mathcal{J}_B = [0, j_B]$ (almost everywhere) boycotts goods up to j_B.*

Comparison with Uniform Import Tariffs

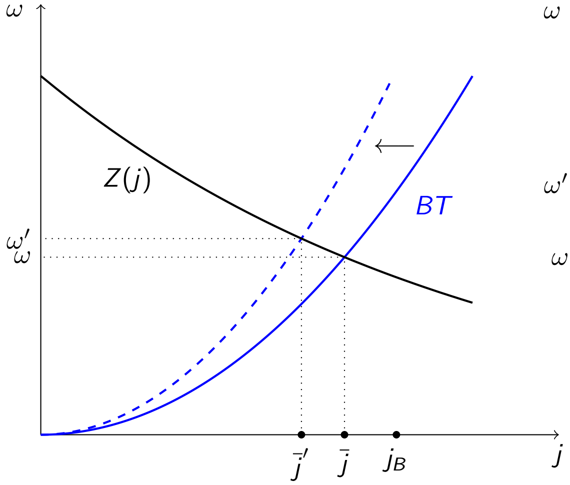

Figure: (A) Equilibrium with Boycott

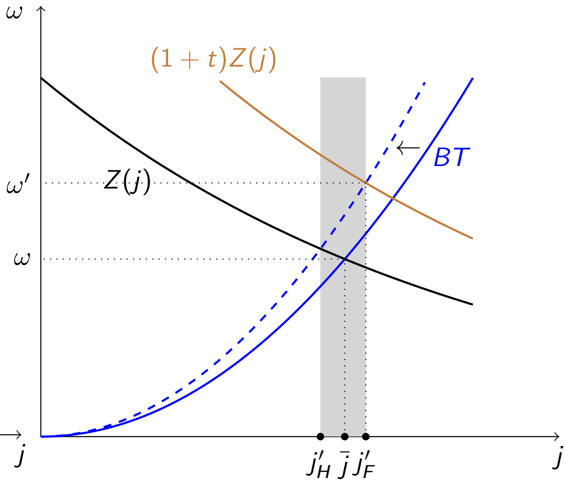

Figure: (B) Linear Import Tariffs

Symmetry

- Consider case with full-boycott-coalition $\lambda = 1$
- In full-boycott-coalition economy, imports of foreign goods are

$$M_H = [1 - S(\mathcal{J}_b)]w_H l_H. \tag{11}$$

- $\uparrow \mathcal{J}_b \implies \uparrow 1 - S(\mathcal{J}_b)$ and indirectly $\uparrow w_H$. On net, $M_H \downarrow$
- Exports of domestic firm goods are

$$X_H = S(\bar{j}(j_B))w_F l_F. \tag{12}$$

- $\uparrow j_B \implies \downarrow \bar{j}(j_B) \implies \downarrow X_H$ Boycotts, therefore, distort both imports and exports.
- Boycott by all consumers is equivalent to a **voluntary export restricitons** by domestic firms.
- Reminiscent of **Lerner symmetry**.

Leakage

- With the presence of nonboycotters $\lambda < 1$, imports from nonboycotters equal to

$$M_{NB} = (1-\lambda)[1 - S(\bar{j}(j_b))]w_H l_H. \tag{13}$$

- Boycott $\uparrow w_H$.
- In addition, the export share $1 - S(\bar{j}(j_b)) \uparrow$ increases when a boycott expands.
- The combination of the two effects raises the imports of nonboycotters M_{NB}
- Related to optimal unilateral behavior by consumers w **altruistic, non-individualistic preferences**.
- Leakage pushes a boycotter to :
  - boycott a larger set of goods to tolerate the same level of Foreign welfare v^F, i.e., $j_B(\lambda, v^F)$,
  - and Foreign imports per boycotter $M_{NB} = [1 - S(j_b)]w_H l_H$ are decreasing in λ.

Labor Supply Side of Boycotts

Modify the DFS setup so that consumers choose consumption goods $\mathbf{q}$ and labor supply l_i to maximize

$$U(\mathbf{q}^i, l_i) = \int_0^1 s(j)\, u\big(q^i(j)\big)\, dj - v(l_i) \tag{14}$$

We consider CRRA utility for goods and an isoelastic disutility of labor:

$$u(q) = \frac{q^{1-\gamma} - 1}{1-\gamma} \quad \text{and} \quad v(l) = \frac{l^{1+\eta}}{1+\eta}, \tag{15}$$

Labor Supply Side of Boycotts

Proposition (Labor Supply of Boycotters)

In every optimal boycott, boycotters work more than regular consumers $l^{B*} > l_H^*$.

Intuition

$$V(w_i, l_i) = \frac{(w_i\, l_i)^{1-\gamma}\, P_i^{\gamma} - 1}{1-\gamma} - \frac{l_i^{1+\eta}}{1+\eta}. \tag{16}$$

where P_i is the price index faced by the consumer. Optimal labor supply of a boycotter

$$l^{B*} = \left[P_B^{\gamma}\, w^{1-\gamma}\right]^{\frac{1}{\eta+\gamma}}.$$

Boycotter faces a penalty in consumption basket bc must more for the goods she buys domestically.

Since $P_B > P$, $l^{B*} > l_H^*$.

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Substitution Elasticities

- Our model extends to cases where the elasticity of substitution $\sigma \neq 1$, adopting CES preferences:

$$U(\mathbf{q}) = \left(\int_0^1 s(j)^{\frac{1}{\sigma}} q^i(j)^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}}$$

- When $\sigma \neq 1$, expenditure on each good depends on its price and the overall price index.
- Results adapt to CES preferences and more general preferences.
- Since consumers find it easier to switch to alternative products with a higher elasticity of substitution relative to that of other goods, it is **optimal to boycott goods close to $\bar{j}$ in relative productivity and substitutable with other goods**

Political Salience and Specific Firm Boycotts

- Consumers may target firms due to political salience or animus, affecting their expenditure shares.
- Preferences and choices for boycotters can include damages $d(j)$ to firms' products:

$$U^B = \max_{\{\mathbf{q}^B\}} \int_0^1 s(j)(1 - d(j)) \log(q^B(j)) dj$$

- If the damage to imported goods is higher, the boycott improves Home's trade terms; otherwise, it harms them.

Strategic Retaliation

- Retaliation involves boycotted countries targeting the home country's exports.
- Strategic interactions can lead to a prisoner's dilemma:

$$\omega = \frac{1-\lambda^F}{1-\lambda^H} \cdot \frac{S(\bar{j})}{1-S(\bar{j})} \cdot \frac{I_F}{I_H}$$

- Utilitarian governments may find no boycotts ($\lambda^H = \lambda^F = 0$) as the only Nash equilibrium.
- So boycotts need to be motivated beyond internal equity considerations.

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Conclusion

- I developed a Ricardian model of boycotts in international trade
- Boycotts can operate on the consumption side or labor supply side
- **Consumption side:** Boycotts can be more effective as ban on goods near comparative advantage barrier.
- **Labor supply side:** Boycotters optimally work more to spend on domestic goods
- Some **symmetry** btw boycotts and VERs but **leakage** from nonboycotters.
- Some open (empirical) questions.
  - How to link measures consumption/labor supply effects to welfare?
  - How large are political salience/other considerations relative to efficiency for boycotts.
  - How large are potential leakages and threats of retaliation?

THANKS!

Comments, complaints, paper boycotts: an1490@nyu.edu